



Generating correlated random vectors

- Straightforward to implement KF steps in Octave.
- To validate KF code via simulation, however, we also must simulate system whose state is being estimated.
 - Produces system input/output data as input to the KF .
 - Provides “true” state value at every point in time.
- But, to simulate the true system, must be able to create nonzero mean Gaussian noise with covariance $\Sigma_{\tilde{y}}$.
- That is, we want $Y \sim \mathcal{N}(\bar{y}, \Sigma_{\tilde{y}})$ but `randn.m` returns $X \sim \mathcal{N}(0, I)$.
 - How to convert X to Y ?



Functions of a random variable

- Suppose we are given two random variables Y and X with $Y = g(X)$. Also assume that g^{-1} exists, g and g^{-1} are continuously differentiable. Then:

$$f_Y(y) = f_X(g^{-1}(y)) \left\| \frac{\partial g^{-1}(y)}{\partial y} \right\|,$$

where $\|\cdot\|$ means to take the absolute value of the determinant.

- So what? Suppose that X is a Gaussian random vector and we form some affine function of X such as:

$$Y = AX + b.$$

- We can now form the pdf of Y $\Rightarrow$ handy!
- This is going to be a key step in learning how to simulate correlated Gaussian RVs.



Example 1 with Gaussian

- Let's find the pdf of $Y = AX + b$ in a couple of steps.
- Start with scalar X and $Y = kX$ and $X \sim \mathcal{N}(0, \sigma_X^2)$.
 - $X = \frac{1}{k}Y$ and so $g^{-1}(y) = \frac{1}{k}y$. Then, $\frac{\partial g^{-1}(y)}{\partial y} = \frac{1}{k}$.

$$\begin{aligned} f_Y(y) &= f_X\left(\frac{y}{k}\right) \left| \frac{1}{k} \right| = \frac{1}{|k|} \frac{1}{\sqrt{2\pi}\sigma_X} \exp\left(\frac{-(y/k)^2}{2\sigma_X^2}\right) \\ &= \frac{1}{\sqrt{2\pi(\sigma_X k)^2}} \exp\left(\frac{-y^2}{2(\sigma_X k)^2}\right). \end{aligned}$$

- Therefore, multiplication by gain k results in a normally-distributed RV with scale change in standard deviation. $Y \sim \mathcal{N}(0, k^2\sigma_X^2)$.



Example 2 with Gaussian

- Now, consider $Y = AX + B$ where A is a constant (non-singular) matrix, B is a constant vector, and $X \sim \mathcal{N}(\bar{x}, \Sigma_{\bar{X}})$.
 - $X = A^{-1}Y - A^{-1}B$ so $g^{-1}(y) = A^{-1}y - A^{-1}B$. Then, $\frac{\partial g^{-1}(y)}{\partial y} = A^{-1}$.
 - $f_X(x) = \frac{1}{(2\pi)^{n/2} |\Sigma_{\bar{X}}|^{1/2}} \exp \left[-\frac{1}{2} (x - \bar{x})^T \Sigma_{\bar{X}}^{-1} (x - \bar{x}) \right]$.
- Therefore,

$$f_Y(y) = |A^{-1}| \frac{1}{(2\pi)^{n/2} |\Sigma_{\bar{X}}|^{1/2}} \exp \left[-\frac{1}{2} (A^{-1}(y - B) - \bar{x})^T \Sigma_{\bar{X}}^{-1} (A^{-1}(y - B) - \bar{x}) \right].$$
- We are going to simplify this expression on the next slide.
 - To do so, note that we can use standard properties of expectation to show that $\bar{y} = A\bar{x} + B$ and $\Sigma_{\bar{Y}} = A\Sigma_{\bar{X}}A^T$.



Example 2 with Gaussian (cont)

- Goal: Simplify the following expression, knowing that $\bar{y} = A\bar{x} + B$ and $\Sigma_{\bar{Y}} = A\Sigma_{\bar{X}}A^T$.

$$\begin{aligned} f_Y(y) &= |A^{-1}| \frac{1}{(2\pi)^{n/2} |\Sigma_{\bar{X}}|^{1/2}} \exp \left[-\frac{1}{2} (A^{-1}(y - B) - \bar{x})^T \Sigma_{\bar{X}}^{-1} (A^{-1}(y - B) - \bar{x}) \right] \\ &= \frac{1}{(2\pi)^{n/2} (|A| |\Sigma_{\bar{X}}| |A^T|)^{1/2}} \exp \left[-\frac{1}{2} (y - \bar{y})^T (A^{-1})^T \Sigma_{\bar{X}}^{-1} A^{-1} (y - \bar{y}) \right] \\ &= \frac{1}{(2\pi)^{n/2} |\Sigma_{\bar{Y}}|^{1/2}} \exp \left[-\frac{1}{2} (y - \bar{y})^T \Sigma_{\bar{Y}}^{-1} (y - \bar{y}) \right]. \end{aligned}$$
- CONCLUSION: Sum of Gaussians is Gaussian—very special case.



Application to our problem

- Suppose that we can find matrix A such that $A^T A = \Sigma_{\bar{Y}}$.
- Then, we generate samples x of random vector $X \sim \mathcal{N}(0, I)$ and compute samples y of random vector $Y = \bar{y} + A^T X$.
 - Since X is zero-mean, $\mathbb{E}[Y] = \mathbb{E}[\bar{y} + A^T X] = \bar{y}$, as desired.
 - The covariance of Y is:

$$\begin{aligned} \mathbb{E}[(Y - \bar{y})(Y - \bar{y})^T] &= \mathbb{E}[(A^T X)(A^T X)^T] \\ &= \mathbb{E}[(A^T X)(X^T A)] \\ &= A^T \underbrace{\mathbb{E}[XX^T]}_I A = \Sigma_{\bar{Y}}, \end{aligned}$$

also as desired.



Important matrix factorizations

- So, if we can find A such that $A^T A = \Sigma_{\tilde{y}}$, we can generate samples x of $X \sim \mathcal{N}(0, I)$ and compute $y = \bar{y} + A^T x$.
- Three ways to generate A :
 - Cholesky factorization of $\Sigma_{\tilde{y}}$ computes A such that $A^T A = \Sigma_{\tilde{y}}$ as long as $\Sigma_{\tilde{y}}$ is positive definite (all eigenvalues are strictly greater than zero).
 - LDL factorization of $\Sigma_{\tilde{y}}$ produces L, D such that $LDL^T = \Sigma_{\tilde{y}}$ and requires only that $\Sigma_{\tilde{y}}$ be positive semi definite: Can then compute $A = \sqrt{D}L^T$.
 - LU factorization of $\Sigma_{\tilde{y}}$ produces L, U such that $LU = \Sigma_{\tilde{y}}$ and requires only that $\Sigma_{\tilde{y}}$ be positive semi definite: Can then compute $A = \text{diag}(\sqrt{\text{diag}(U)})L^T$.
- Cholesky and LU are built into Octave; all are built into MATLAB.



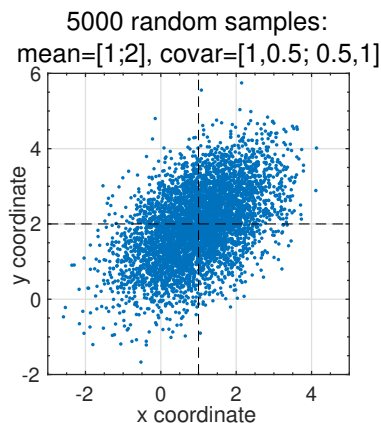
Example Octave code

- Suppose we wish to generate a single sample of y using Cholesky method:

```
ybar = [1; 2];
covar = [1, 0.5; 0.5, 1];
A = chol(covar);
x = randn([2, 1]);
y = ybar + A'*x;
```

- Suppose we wish to generate 5000 samples of y using the LU method:

```
ybar = [1; 2];
covar = [1, 0.5; 0.5, 1];
[L,U] = lu(covar);
A = diag(sqrt(diag(U)))*L';
x = randn([2, 5000]);
y = ybar(:,ones([1 5000]))+A'*x;
plot(y(1,:),y(2,:), 'b');;
```



Summary

- In order to test KF code via simulation, must also simulate system whose state is being estimated.
- This requires ability to simulate (possibly) correlated (possibly nonzero mean) noises.
- Octave natively produces samples of zero-mean uncorrelated Gaussians only.
- But, can transform these to have desired mean and correlation using a matrix determined using either the Cholesky or LU matrix decompositions, both of which are built into Octave.